

Three-Level Audit of Normative-Frame Sensitivity in LLM-Assisted Policy Simulation

Model-level

H1: Composite orientation

$$\text{MFQ}_m \rightarrow \bar{\Delta}_m$$

Does aggregate moral-foundations orientation predict cross-model burden effects?

N = 15 model endpoints

$$\beta_1 = 2.96, 95\% \text{ CI } [-2.17, 8.09]$$

$$R^2 = 0.11$$

No aggregate signal

Narrative-level

H2: Narrative frame association

$$\text{Frame}_{mir} \rightarrow \Delta_{mir}$$

Do naturally expressed rationale frames track simulated burden effects?

Model *imes* profile pair

fixed effects

N = 2,019

Access barriers: $b = 0.13$, n.s.

Coercive backlash: $b = 1.21$, n.s.

Collective resp.: not estimable (1/2,019)

No robust within-pair signal

Within-model

H3: Frame manipulation

$$\text{Assigned Frame}_c \rightarrow \Delta_{mcir}$$

Does an assigned prompt-frame clause shift the simulated burden effect within the same model?

4 models *imes* 27 profiles

imes 4 frames

Autonomy: +4.96

Equity/access: +4.84

Collective obligation: -10.88

Frame-sensitive

Model-specific heterogeneity

Aggregate orientation: no signal · Natural rationale frames: no robust within-pair signal

Assigned framing: shifts simulated effects